CONCEPT

**Tech Stack**

| Layer | Technology | Purpose |
|---|---|---|
| Orchestration | Apache Airflow 3.1.8 | Pipeline scheduling & monitoring |
| Containerization | Docker | Isolated runtime environment |
| AI Layer | Google Gemini 1.5 Flash | Sentiment analysis & summarization |
| Data Sources | BBC, Reuters, Sky News RSS | Live world news feeds |
| Primary Storage | SQLite | Structured queryable database |
| Export Format | CSV | Portable data export |
| Language | Python 3.12 | Pipeline logic |
| Data Parsing | XML ElementTree | RSS feed parsing |
| HTTP Client | Requests | API & feed calls |

Every hour Airflow scheduler triggers the pipeline:

SCHEDULING & AUTOMATION

00:00 ⟶ Pipeline starts

00:01 ⟶ News fetched from 3 sources (~50 articles)

00:02 ⟶ Data cleaned and deduplicated (~45 articles)

00:03 ⟶ Top 10 articles analyzed by Gemini AI

00:04 ⟶ Results saved to CSV and SQLite

00:05 ⟶ AI daily briefing report generated

Pipeline complete ✅

Next run: 01:00 ──▶ repeats automatically

```
**SYSTEM ARCHITECTURE**

┌──────────────────────────────────────────────────────────────┐
│                    APACHE AIRFLOW (Docker)                   │
│                                                              │
│  ┌─────────┐    ┌─────────┐    ┌─────────┐    ┌─────────┐   │
│  │ FETCH   │──▶│ CLEAN   │──▶│ AI      │──▶│ SAVE    │   │
│  │ NEWS    │    │ FILTER  │    │ ANALYZE │    │ DATA    │   │
│  └─────────┘    └─────────┘    └─────────┘    └─────────┘   │
│       │                             │              │        │
│  3 RSS Feeds                Gemini AI API      ┌──▼──┐      │
│  BBC, Reuters,              Summary +          │ CSV │      │
│  Sky News                   Sentiment +        │SQLite│     │
│                             Category           └─────┘      │
│                                                   │         │
│                                              ┌──▼──┐        │
│                                              │AI DAILY│     │
│                                              │REPORT │      │
│                                              └─────┘        │
└──────────────────────────────────────────────────────────────┘
```

## 🔄 Pipeline Flow — Step by Step

### Task 1 — fetch_news

```
BBC World RSS ──┐
Reuters RSS ────┼──▶ Parse XML ──▶ Extract Fields ──▶ XCom Push
Sky News RSS ───┘

Fields extracted:
- title
- description
- link
- published date
- source
- fetched timestamp
```

```
**Task 2 – clean_news**
Raw Articles ──▶ Remove Duplicates ──▶ Remove Empty Titles
                      ──▶ Strip HTML Tags
                      ──▶ Decode Special Characters
                      ──▶ Clean Articles ──▶ XCom Push
**TASK 3 - AI-analyze**
Clean Articles ──▶ For each article:
                      |
                      ▼
                 Call Gemini API
                      |
                      ▼
                 Returns JSON:
                 {
                   "summary": "...",
                   "sentiment": "positive/negative/neutral",
                   "sentiment_reason": "...",
                   "category": "politics/economy/..."
                 }
                      |
                      ▼
                 Analyzed Articles ──▶ XCom Push
```

```
**Task 4 — save_to_csv**

Analyzed Articles ──▶ Check if file exists
                              │
                         ┌────▼────┐
                         │  New?   │
                         └─────────┘
                              │
                     Yes ─────┴───── No
                      │               │
                Write Header    Append Only
                      │               │
                      └───────┬───────┘
                              ▼
                        world_news.csv

**Task 5 — save_to_sqlite**

Analyzed Articles ──▶ Connect to SQLite
                              │
                              ▼
                       CREATE TABLE IF
                       NOT EXISTS ai_news
                              │
                              ▼
                       INSERT each article
                              │
                              ▼
                       Commit & Close
```

```
**Task 6 – ai_daily_report**

Analyzed Articles ──▶ Count sentiments
                          |
                          ▼
                    Build headlines text
                          |
                          ▼
                    Call Gemini API
                          |
                          ▼
                    AI writes 3-4 sentence
                    professional briefing
                          |
                          ▼
                    Print full report
                    to Airflow logs
```

## 📁 Project Structure

```
airflow-test/
├── docker-compose.yaml        # Airflow Docker setup
├── .env                       # Environment variables
├── dags/
|   ├── weather_pipeline.py    # Project 1 - Weather Pipeline
|   └── ai_news_pipeline.py    # Project 2 - AI News Pipeline
├── logs/
|   ├── ai_news.csv            # News data export
|   └── ai_news.db             # SQLite database
└── plugins/                   # Airflow plugins
```